

Identification and Classification of Encrypted Network Traffic Flows

Systems and Network Security — Project Report

Tony Vukovic · Arij Kristou · Aurore Bellessa

Context & Motivation

Encryption is the new default

TLS 1.3, QUIC, DoH — the majority of internet traffic is now encrypted

DPI is no longer viable

Traditional Deep Packet Inspection cannot inspect encrypted payloads

ML as a solution

Flow-level metadata (timing, packet sizes, window init) enables payload-free classification

Research Question

To what extent can statistical flow metadata alone support reliable, privacy-preserving threat detection in fully encrypted network environments — and how does the choice of extraction tooling affect that capacity?

Related Work

Classification

- Flow metrics suffice without payload access (Moore & Papagiannaki, 2005)
- JA3/JA3S TLS fingerprinting
- QUIC metadata & inter-packet timing
- CNN-based encrypted classifiers (Okonkwo et al., 2022)

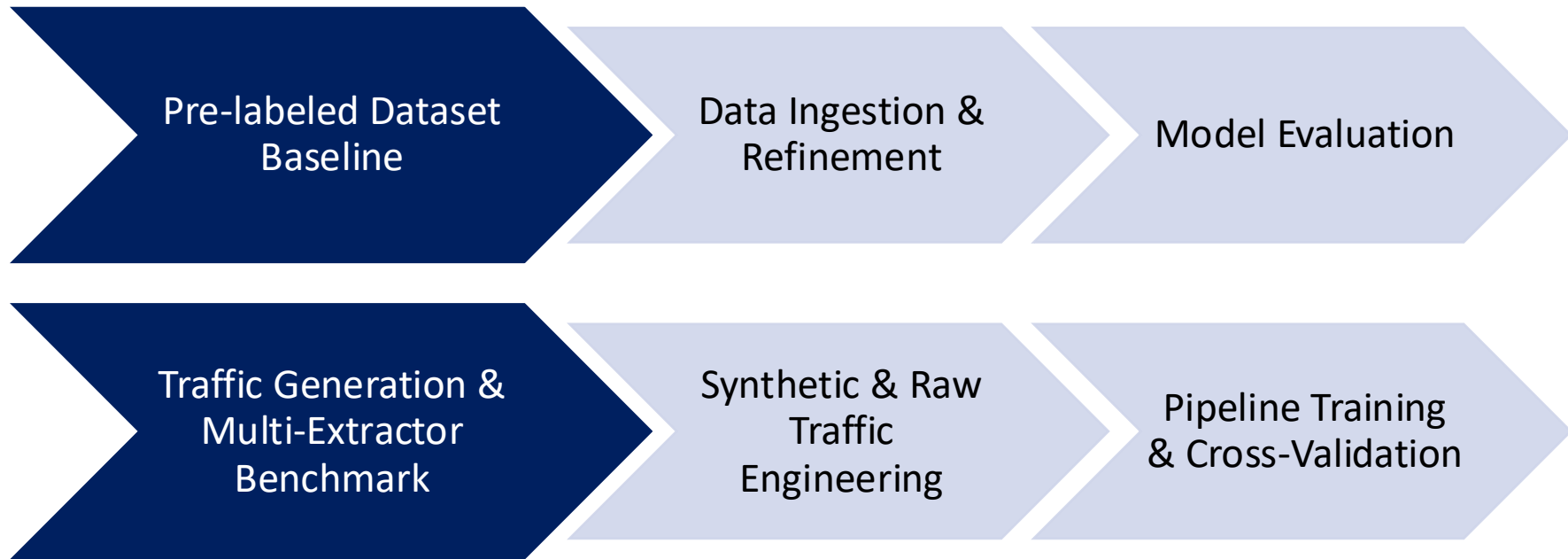
Extraction Tools

- NFStream — Python-native, rich flow stats
- Cisco Joy — TLS/SSH handshake telemetry
- PcapPlusPlus — C++, high-speed capture
- PyShark / Scapy — ground-truth validation

Benchmark Datasets

- CIC-IDS2017, CSE-CIC-IDS2018, CIC-DDoS2019
- Known limitation: severe class imbalance
- CIC-IDS-Collection (Kaggle): harmonized labels across all datasets

Plan of the study



Dataset & Encrypted Traffic Isolation

9.1M

Total flow records

59

Statistical features

4.75M

Encrypted flows retained

8

Attack categories

Behavioral Fingerprinting Heuristic

Isolates bidirectional handshake patterns characteristic of TLS and SSH sessions.
Port numbers and IP excluded deliberately to prevent overfitting.

- $\text{Init_Fwd_Win_Bytes} > 0$ — positive forward window size
- $\text{Init_Bwd_Win_Bytes} > 0$ — positive backward window size
- $\text{Fwd_Seg_Size_Min} \geq 20$ — minimum forward segment length

8 Attack Categories:

Benign

DDoS

DoS

Botnet

Bruteforce

Infiltration

Webattack

Portscan

Issues

Class unbalanced

Severe baseline asymmetry. Benign network traffic dominates the dataset, reflecting real environment.

Trivial Classification shortcut. Standard ML models bias heavily toward the majority class, leading to high global accuracy but a critical blindness toward active threats.

Infiltration attack

Statistical indistinguishability of infiltration threats. Their behaviors are heavily overlapping with legitimate web features within highly encrypted domains.

Classification Pipelines

1

Baseline Random Forest

50 estimators · max depth
15
Stratified 70/30 split ·
StandardScaler

2

Custom Feature

Flow_Repetition_Count:
flows sharing identical
mean & max forward
packet length

3

MLP Neural Network

2 hidden layers (50 + 25
neurons)
Early stopping · max 30
iterations

4

SMOTE Oversampling

Synthetic minority
oversampling targeting
rare attack classes

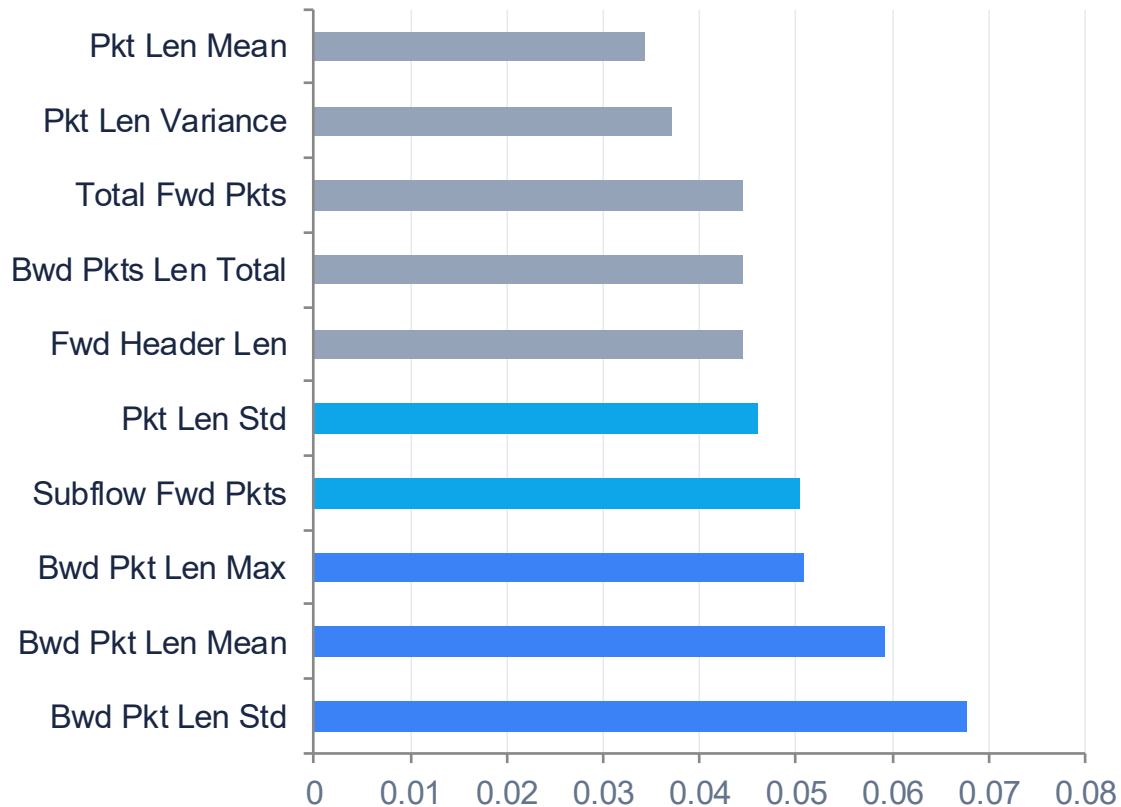
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Cost-Sensitive RF

class_weight='balanced'
Prioritizes sensitivity over
global accuracy

Goal: identify the best strategy for detecting stealthy attacks like Infiltration in encrypted flows

Feature Importance Results



Backward stats dominate — server response patterns are the primary fingerprint

Packet shape > volume — distribution matters more than total bytes

Flow_Repetition_Count absent from top 15 — repetition-based signatures add no signal

Model Comparison Results

Classification report by class					
	precision	recall	f1-score	support	
Benign	0.99	1.00	0.99	1128433	
Botnet	1.00	1.00	1.00	43180	
Bruteforce	1.00	1.00	1.00	30363	
DDoS	1.00	1.00	1.00	160561	
DoS	1.00	1.00	1.00	51115	
Infiltration	0.72	0.02	0.04	12799	←
Portscan	1.00	0.95	0.97	363	
Webattack	1.00	0.91	0.95	770	
accuracy			0.99	1427584	
macro avg	0.96	0.86	0.87	1427584	
weighted avg	0.99	0.99	0.99	1427584	

Model Comparison Results

Model	Accuracy	Inf. Recall	Inf. Precision	Inf. F1
Baseline RF	~99%	≈ 0	≈ 0	≈ 0
RF + SMOTE	99.00%	0.03	0.60	0.06
RF + Class Weight	82.51%	0.93	0.05	0.09
MLP Neural Network	98.45%	0.11	0.17	0.13

Volumetric attacks (DDoS, DoS, Botnet, Bruteforce) → near-perfect detection across all models

Sensitivity vs. reliability trade-off: high recall = high false-positive rate (RF + Class Weight)

Infiltration remains unresolved — not a model problem, but a feature-space overlap problem

Benchmark Pipeline Overview

From raw captured traffic to classified flows — three extraction tools, one ML pipeline



Each extraction tool feeds an independent ML pipeline — results are compared across tools to assess structural impact on classification

Flow Extraction Tools

NFStream

91%

Leakage

- Python-focused, high-performance live and pcap processing framework.
- Rich statistical metadata (87 features)
- Flow stats overshadowed by app metadata

PcapPlusPlus

78%

Insufficient

- Multiplatform C++ engine for low-level packet parsing and analysis.
- Extremely fast execution with minimal memory footprint, directly wrapping
- Only 2 features extracted
- Designed for capture speed, not statistical analysis

Cisco Joy

79%

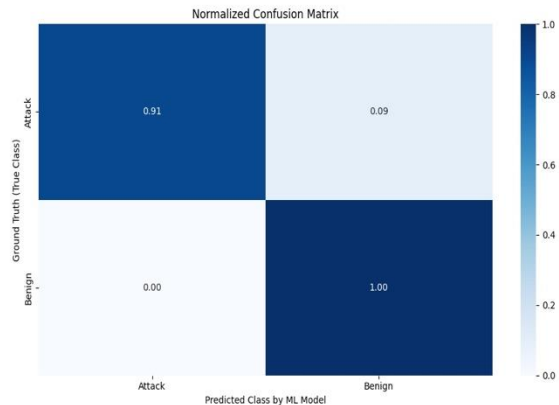
Binary only

- Industry-standard enterprise network telemetry
- Produces lightweight, highly standardized flow records optimized for high-throughput production environment
- Binary schema: Attack vs. Benign only
- No per-category analysis possible

Confusion Matrices — Extraction Tool Benchmark

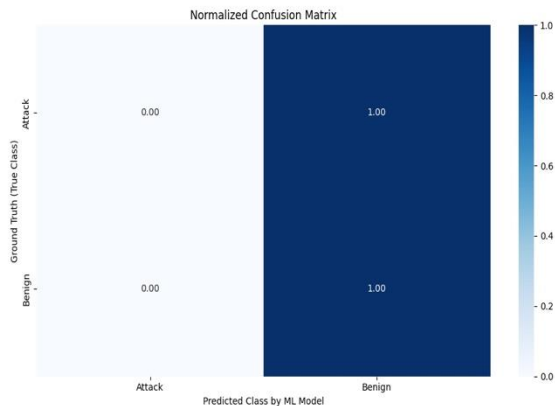
Normalized per true class

NFStream — Binary (Attack vs. Benign)



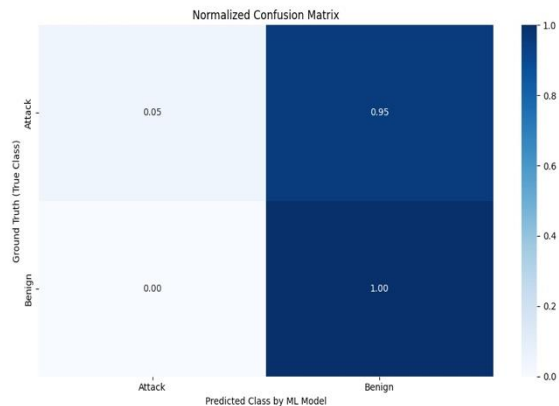
98%

PcapPlusPlus — Binary (Attack vs. Benign)



78%

Cisco Joy — Binary (Attack vs. Benign)



79%

Discussion

No single tool is sufficient

NFStream offers rich statistics but use future leakage. PcapPlusPlus is operationally fast but statistically sparse. Cisco Joy is also fast but produces mixed results and forces binary classification.

Infiltration is a representation problem

Neither SMOTE, cost-sensitive learning, nor a neural architecture resolves Infiltration detection. The failure is in the feature space — stealthy flows are statistically indistinguishable from benign encrypted traffic without richer protocol-level telemetry.

Toward a multi-tool production pipeline

Combining NFStream's statistical depth (with application metadata excluded) and Cisco Joy's handshake fingerprinting, while reserving PcapPlusPlus for high-speed capture, represents the most promising direction.

Conclusion

Answer to the research question

Flow-level metadata reliably detects volumetric encrypted attacks, but is fundamentally insufficient for low-volume stealthy intrusions without richer protocol-level telemetry.

DDoS, DoS, Botnet, Bruteforce → near-perfect detection with flow metadata alone

Infiltration → unresolved across all models and balancing strategies

Tool choice is not neutral — NFStream, PcapPlusPlus, and Cisco Joy each impose structural limitations

Future work: live NFStream integration + multi-tool pipeline + temporal behavioral modeling